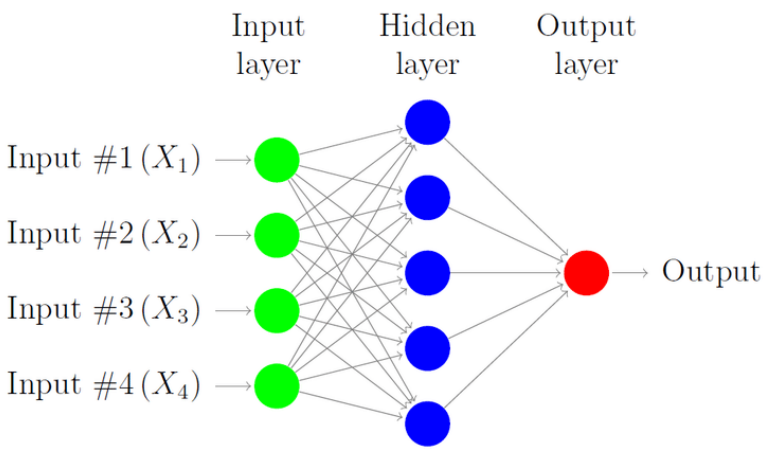
One Layer Functional Link Nueral Network Controller



$y = W^T \phi(x)$

x : **input vector**

y : **output vector**

W : **weight matrix**

$\phi(x)$: **main function**

$\tau = M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) + F(\dot{q}) + \tau_d$

$r = \dot{e} + \lambda e = (\dot{q} - \dot{q}_d) + \lambda(q - q_d)$

$\dot{r} = \ddot{e} + \lambda \dot{e} = (\ddot{q} - \ddot{q}_d) + \lambda \dot{e}$

$\ddot{q} = \dot{r} + \ddot{q}_d - \lambda \dot{e}$

$M\dot{r} = -M(\ddot{q}_d - \lambda \dot{e}) - C\dot{q} - G - F - \tau_d + \tau$

- **Some of the** $C * \dot{q}$ **and friction terms are included in** $-V_m * r$
- $f(x)$: **The part of** $-M(\ddot{q}_d - \lambda * \dot{e}) - C * \dot{q} - G - F$ **not included in** $-V_m * r$

$M\dot{r} = -V_m r + f(x) + \tau_d - \tau$

$-V_m r$: **damping element**

$+f(x)$: **unknown load**

$+\tau_d$: **disruptive effect**

$-\tau$: **controller torque**

$\tau = \hat{f} + K_v r - v(t)$

$=> M\dot{r} = -V_m r + f(x) + \tau_d - (\hat{f} + K_v r - v(t))$

$=> M\dot{r} = -V_m r + \tilde{f} - K_v r + \tau_d + v(t)$

$y = W^T \phi(x) => f(x) = W^T \phi(x) + \varepsilon \quad \hat{f} = \hat{W}^T \phi(x)$

$\tau = \hat{W}^T \phi(x) + K_v r - v$

$\tilde{W} = W - \hat{W}$

$=> f - \hat{f} = W^T \phi(x) + \varepsilon - \hat{W}^T \phi(x) = (W - \hat{W})^T + \varepsilon = \tilde{W}^T \phi(x) + \varepsilon = \tilde{f}$

$M\dot{r} = -V_m r + \tilde{W}^T \phi(x) + \varepsilon - K_v r + \tau_d + v = -(V_m + K_v) r + \hat{W}^T \phi(x) + \varepsilon + \tau_d + v$

$\varepsilon + \tau_d + v -> $ **neglected**

$M\dot{r} = -(V_m + K_v) r + \hat{W}^T \phi(x)$

$L = \frac{1}{2} r^T M r + \frac{1}{2} tr\left\{\tilde{W}^T F^{-1} \tilde{W}\right\}$ **- > Lyapunov function candidate**

$L_1 = \frac{1}{2} r^T M r \quad L_2 = \frac{1}{2} tr\left\{\tilde{W}^T F^{-1} \tilde{W}\right\}$

$\dot{L}_1 = \frac{d}{dt}(\frac{1}{2} r^T M r) = \frac{1}{2}\left(\frac{d}{dt}(r^T) M r + r^T \frac{d}{dt}(M) r + r^T M \frac{d}{dt}(r)\right)$

$M = M^T, \ \dot{r}^T M r = (r^T M \dot{r})^T = r^T M \dot{r}$

$= \frac{1}{2}\left(\dot{r}^T M r + r^T \dot{M} r + r^T M \dot{r}\right) = r^T M \dot{r} + \frac{1}{2} r^T \dot{M} r$

F:constant

$\dot{L}_2 = \frac{1}{2} tr\left\{\dot{\tilde{W}}^T F^{-1} \tilde{W} + \tilde{W}^T F^{-1} \dot{\tilde{W}}\right\}$

$\dot{L}_2 = tr\left\{\tilde{W}^T F^{-1} \dot{\tilde{W}}\right\}$

$r^T (M\dot{r}) = r^T (-(K_v + V_m) r + \tilde{W}^T \phi(x))$

$r^T M\dot{r} = -r^T K_v r - r^T V_m r + r^T \tilde{W}^T \phi(x)$

$\dot{L} = \dot{L}_1 + \dot{L}_2 = r^T M\dot{r} + \frac{1}{2} r^T \dot{M} r + tr\left\{\tilde{W}^T F^{-1} \dot{\tilde{W}}\right\}$

$\dot{L} = -r^T K_v r - r^T V_m r + r^T \tilde{W}^T \phi(x) + \frac{1}{2} r^T \dot{M} r + tr\left\{\tilde{W}^T F^{-1} \dot{\tilde{W}}\right\}$

$\dot{M} - 2V_m = S(q) \quad S(q)$ **: skew-symmetric matrix**

$=> r^T (\dot{M} - 2V_m) r = 0 -> $ **skew-symmetric property**

$r^T \dot{M} r - -2V_m r^T r = 0 => \ \frac{1}{2} r^T \dot{M} r = r^T V_m r$

$\dot{L} = -r^T K_v r + (-r^T V_m r + \frac{1}{2} r^T \dot{M} r) + r^T \tilde{W}^T \phi(x) + tr\left\{\tilde{W}^T F^{-1} \dot{\tilde{W}}\right\}$

$= -r^T K_v r + r^T \tilde{W}^T \phi(x) + tr\left\{\tilde{W}^T F^{-1} \dot{\tilde{W}}\right\}$

$r^T \tilde{W}^T \phi(x) = tr\left\{\tilde{W}^T \phi(x) r^T\right\}$

$\dot{L} = -r^T K_v r + tr\left\{\tilde{W}^T (F^{-1} \dot{\tilde{W}} + \phi(x) r^T)\right\}$

• **in order to ensure stability** $\dot{L} \le 0$

$$tr\left\{\tilde{W}^T(F^{-1}\dot{\tilde{W}} + \phi r^T)\right\} = 0$$

$$F^{-1}\dot{\tilde{W}} + \phi r^T = 0 \;\; => \;\; F^{-1}\dot{\tilde{W}} = -\phi r^T$$

$$\dot{\tilde{W}} = -F\phi r^T$$

$$\dot{\tilde{W}} = -\dot{\hat{W}} \;\; => \;\; \dot{\hat{W}} = F\phi r^T$$

$$\dot{L} = -r^T K_v r$$

$$\lim_{t\to\infty} \dot{L} = 0 \;\; \to \;\; \lim_{t\to\infty}(-r^T K_v r) = 0 \quad \to \lim_{t\to\infty} r^T = 0$$

•$r(t)$ **goes to zero over time, which means that** $e(t) = 0$ **and** $\dot{e}(t) = 0$**, indicating the control is successful.**

control input : $\tau = \hat{W}^T \phi(x) + K_v r$

tuning algorithims : $\dot{\hat{W}} = F\phi r^T \quad F > 0$